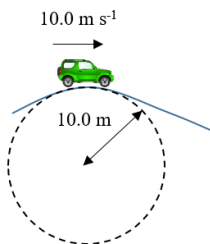


Q9



When the car is moving on a level road,
tension in spring = weight of mass

$$T = mg$$

$$ke_0 = mg \quad - (1)$$

where e_0 = equilibrium extension of spring.

When passing over a hump, the centripetal acceleration is downwards, so the net force is downwards. Hence, the spring shortens.

Centripetal force (downwards) = $mg - T_1$

$$\frac{mv^2}{r} = mg - T_1$$

$$\frac{mv^2}{r} = mg - ke_1$$

Using (1), we get:

$$\frac{mv^2}{r} = ke_0 - ke_1$$

$\therefore$ the decrease in the extension of the spring is:

$$e_0 - e_1 = \frac{mv^2}{kr}$$

$$e_0 - e_1 = \frac{0.250 \times 10^2}{25.0 \times 10.0} = 0.100 \text{ m} = 10.0 \text{ cm}$$

So, the spring shortens by 10.0 cm.

Ans: A

